Response to referee-reports

Manuscript: Generic deformation channels for critical Fermi surfaces including the impact of collisions

Dear Editor,

Thank you very much for arranging to review our manuscript. We have addressed all the questions raised by the referee. Detailed responses to the queries of the referee have been provided below (Referee's comments are highlighted in blue). In view of these changes, we trust the manuscript will now be judged suitable for publication.

Sincerely,
Authors

REVIEWER 1

1. In the manuscript, the authors assume the existence of a stable non-Fermi-liquid (nFL) fixed point in the low-energy limit. However, for Ising-nematic quantum criticality, the existence of such a stable nFL fixed point remains unsettled due to the lack of a controlled analytical approach. For example, in the large-N framework, Lee [3] has shown that even in the large-N limit, all planar diagrams contribute in the low-energy regime and must be taken into account. To the best of my knowledge, there are no definitive results establishing a stable NFL fixed point in this context. Given this situation, it is unclear how the use of fermionic and bosonic Green's functions incorporating only a limited number of loop corrections can be justified. Could the authors clarify the assumptions underlying this approximation? Alternatively, is it possible to relate their approach to recently proposed SYK-based models, such as that of Esterlis et al. [1], where a controlled non-Fermi-liquid fixed point is more firmly established?

   The existence of a stable NFl fixed point is now a settled problem thanks to the controlled approximation showed in papers cited as Refs. [1, 2], using dimensional regularization of the QFT framework. Hence, a fully controlled and rigorously established NFL fixed point at the Ising-nematic QCP no longer remains an open theoretical problem since 2014. We urge the referee to read these papers and also the review by S. S. Lee [3] explaining these developments. Our work is based on this stable NFL is therefore completely justified. Our fermionic Green's function and one-loop corrected bosonic Green's function are taken from those studies and, in particular, the $(2+1)$-d case studied in Ref. [1]. Due to the controlled nature of the expansion serious, one can choose to work at the leading-order correction involving the one-loop order. We also refer the referee to the first paper in this series, viz. Ref. [4], where a clear derivation of the starting action in the Keldysh formalism is shown, starting with the equilibrium action the stable NFL fixed point at the Ising nematic QCP. In particular, Sec. II of Refs. [4, 5] contains all these details. All the above references have been properly cited in our manuscript and the origins of our action appropriately explained in Sec. II (entitled "Model") of our manuscript, without repeating every detail explained already in Refs. [4, 5].

   The SYK-based models are not applicable to our studies reported here.

2. The manuscript appears to use incorrect definitions of momenta conjugate to relative and center-of-mass coordinates. The correct definitions should be $q = k_1 - k_2$ and $k = (k_1 + k_2)/2$.

   Upon re-examination, we find that the Wigner transformation and subsequent derivations of the QBEs are performed correctly and have been verified with the derivations presented in Refs. [4–7]. To make a clear demonstration of how the relative and total momentum are related to the relative and center-of-mass coordinates in the Wigner transformation, let us assume that we need to perform the following operation $A \circ B$, where $A(x_1, x_2), B(x_1, x_2)$ are some two-point function of co-ordinates $x_{1,2} = (t_{1,2}, \mathbf{r}_{1,2})$, and the operation $\circ$ represents

$$\begin{aligned} A \circ B &= \int dx'\, A(x_1, x')\, B(x', x_2) \\ &= \int_{k, k_2, k_3, k_4} A(k_1, k_2)\, B(k_3, k_4)\, e^{i\left(k_1.x_1 + k_2.x' + k_3.x' + k_4.x_2\right)}, \end{aligned} \tag{1}$$

   where in the second line, we perform Fourier transformation. Now, we write $x_1 = x + x_{\rm rel}/2$ and $x_2 = x - x_{\rm rel}/2$, where $x = (x_1 + x_2)/2$ is the center-of-mass coordinate and $x_{\rm rel} = x_1 - x_2$ is the relative co-ordinate, and find, after performing the space and momentum integration,

$$\begin{aligned} A \circ B &= \int_{k_1, k_2, k_4} A(k_1, k_2)\, B(-k_2, k_4)\, e^{i(k_1 + k_4).x}\, e^{i(k_1 - k_4).x_{\rm rel}/2}, \\ &= \int_{q, k, k_2} A(q + \frac{k}{2}, k_2)\, B(k_2, q - \frac{k}{2})\, e^{iq.x}\, e^{ik.x_{\rm rel}}, \end{aligned} \tag{2}$$

where we define the relative momentum $k = (k_1 - k_4)/2$ and total momenta $q = (k_1 + k_4)$. This shows that, if the Wigner coordinates are defined as (we follow this in Sec. IIA)

$$x_{\rm rel} = x_1 - x_2, \qquad x = \frac{x_1 + x_2}{2},$$

then the canonically conjugate variables are

$$k = \frac{k_1 - k_2}{2}, \qquad q = k_1 + k_2.$$

3. There have been numerous previous studies of non-Fermi-liquid states using Boltzmann or kinetic-equation approaches. For example, related frameworks have been developed based on bosonization [2] and SYK- based models [1]. In this context, it would be beneficial for the authors to compare their results with those obtained in earlier works. In particular, a clearer discussion of what is genuinely new in the present manuscript would strengthen its contribution. It would also be helpful to comment on the relative strengths and limitations of the current approach in comparison with existing methods in the literature, and to clarify in which regimes the present formalism offers advantages or faces potential shortcomings.

We believe we already addressed these important points in our original manuscript. In the "Discussions, summary, and future outlook" section, we had already discussed the wider applicability of our new results, compared to the approximations adopted in our earlier works [4, 5]. We repeat it here for the convenience of the referee:
"In this paper, we have analyzed the nature of generic displacements of the critical Fermi surface for the NFL system arising at the Ising-nematic QCP, by using the quantum-Boltzmann-equation formalism applicable for our NFL scenario. By incorporating a generalized fermionic distribution function, we have derived the self-consistent equations whose solutions give us the functional forms of the Fermi-surface displacements as well as their dispersions. In particular, we have been focused on determining the dispersion and stability of the low-energy and low-momentum excitations/collective modes, arising from these displacements. Compared with our earlier work, we have included here the effects of damping, originating from the collision integral for the fermionic QBE. We have discussed the implications in two scenarios: (1) The simplistic F_0 model, where we are able to find an analytical solution by simply setting $F(\theta, \Omega) = F_0(\Omega)$, retaining only the $\ell = 0$ angular-momentum component. (2) The F_ℓ model, where generic $\ell \neq 0$ components are included, which warrants a numerical simulation to obtain the explicit results. A summary of our findings is elucidated below:

(a) Within the F_0 model, the dispersion of the zero sound of our NFL exhibits a $\Omega_r \sim |\mathbf{q}|^{6/5}$ behaviour at the lowest energy scales, unlike the $\Omega_r \sim |\mathbf{q}|$ dependence predicted from the Fermi-liquid theory. This fractional-power-law dependence is seen to morph into an $\Omega_r \sim |\mathbf{q}|$ behaviour when the energy is cranked up above a crossover scale, Ω_0. As a consequence of accounting for the collision terms, the dispersion acquires a nonzero imaginary part, Ω_i, which corresponds to damping. However, $|\Omega_i|$ is parametrically much smaller than Ω_r, making the $\ell = 0$ mode a long-lived excitation. Ω_i is also seen to cross over to distinct power-law characteristics with respect to Ω_0. As a function of the mode's momentum, the ratio of $|\Omega_i/\Omega_r|$ decays rapidly in a monotonic way (in all the applicable energy ranges), and becomes more and more negligible.

(b) Incorporating the F_ℓ model, for very small momenta, the nature of the zero sound is found to have linear dependence on momentum and a negligible imaginary part. Intriguingly, in addition to the zero-sound solution, many new collective-mode excitations of the critical Fermi surface emerge, whose Ω-values lie between those of the zero sound and particle-hole continuum (when visualised in the complex Ω-plane). At extremely-low momenta, the number of these additional modes proliferate [cf. Fig. 5(b)]. However, with increasing $\bar{q}$, the extra islands disappear progressively and the overall behaviour crosses over to the results obtained from the F_0 model.

We have extended our calculations to include the effects of the bosonic density fluctuations as well. Interestingly, the $T = 0$ results for the Fermi-surface displacements remain unchanged under this generalisation."

Regarding the comparision with other kinds of NFLs having different origins, the situation we have to keep in mind that "All happy families are alike; each unhappy family is unhappy in its own way." In our context, "Happy" refers to "Fermi liquids" and "Unhappy" refers to "NFLs". Hence, it does not make very much sense to compare the study involving an NFL arising at Ising-nematic QCP possessing with the NFLs having different microscopic origins like SYK-based models. Nevetheless, we have now added the following discussions:
"

(a) Relation to earlier Boltzmann and kinetic approaches:
Traditional Boltzmann or kinetic-equation treatments of quantum critical metals typically rely either on the existence of long-lived quasiparticles or focus on the collisionless regime. In contrast, our analysis formulates quantum Boltzmann equations directly in terms of Keldysh Green's functions and does not assume quasiparticle-coherence in the frequency space [4, 6, 7]. The key new element relative to earlier treatments is the systematic inclusion

of *collision-induced damping* arising from the imaginary parts of the GLFs. This allows us to analyze not only collective-mode dispersions but also their intrinsic damping across generic angular-momentum channels at finite momenta ($\mathbf{q}$) and frequencies ($\mathbf{\Omega}$).

(b) Relation to bosonization-based approaches:
Recent bosonization-based treatments of NFL states [8] emphasise on kinematic constraints and operator-level formulations of Fermi-surface fluctuations. While these methods provide valuable insights into universal scaling properties, they typically do not incorporate collision integrals or damping effects in a transparent kinetic form. Our approach is complementary: it retains explicit momentum resolution along the Fermi surface and provides a direct kinetic description of collective modes, including damping, which is not readily accessible in current bosonization frameworks.

(c) Relation to SYK-based models:
SYK-inspired models are based on large-N formulation (e.g., Ref. [9]), and depend crucially on the large-N limit. Their microscopic formulation is fundamentally different form working the QFT-based action adopted here. We take the leading order fermion-boson interaction descibing the Ising-Nematic QCP and the controlled approximation dealing with this strongly-couple theory is well-established within the dimensional-regularization scheme put forward in Refs. [1, 2]. Our QFT analysis has no dependence whatsoever on the number of flavours (N) of fermions and its validity is not contingent upon the large-N limit. In particular, we have not found an exhaustive analysis an exhaustive analysis of the collective modes of critical Fermi surfaces including the full effects of the collision integrals like we have done, and neither is there a full explanation of the emergent collective modes quantified by their structure as well as (Ω, $\mathbf{q}$)-dependence. In other words, our QBE framework is designed to address all these aspects in aholisitc way by incorporating the spatial structure, Fermi-surface geometry, energy-dependence, and finite-$\mathbf{q}$ dynamics, which are currently lacking in those SYK-based studies.

(d) Novelty and limitations:
The principal new contributions of our work, beyond earlier works, can be summarised as

- inclusion of frequency-dependent *complex* GLFs encoding collision-induced damping;
- systematic analysis of collective modes in generic angular-momentum channels;
- extension of the QBE framework to finite (Ω, q) beyond the collisionless limit. Limitations, if any, can be attributed to carrying out our analysis in the $T \to 0$ regime.

In the "Introduction" section, we have added the following lines:
"Overall, we would like to emphasise that the primary new advancement of our work is twofold: (1) the systematic inclusion of *collision integral part* within the QBE framework, going beyond the collisionless analysis of earlier works (viz. Refs. [4, 5]); and (2) studying the frequency-dependent and complex GLFs, encoding damping effects to give a full-fledged description of generic deformation channels for critical Fermi surfaces."

Overall, we would like to emphasise that our approach is best viewed as an effective low-energy description, conditional on the existence of an NFL scaling regime, and is built on the stable fixed point obtained by the method of dimensional regularization in a *strictly controlled approximaiton*.

REVIEWER 2

1. There is a high fraction of self-citations, particularly to the authors' own earlier papers that develop the QBE framework and the collisionless analysis. While these citations are often unavoidable, the reference list could be broadened to better situate the work within the wider literature on quantum critical metals beyond the author's framework.

We have now added discussions and more citations highlighting the wider literature studying NFLs. In particular, they now include bosonization-based methods [8], SYK-based models [9], and a review of controlled approximation based on dimensional regularization [3, 10]. We have added the following lines:
"There have been extensive efforts to obtain perturbative NFL states implementing distinct and complementary ways, for which we provide an overview. The first popular attempt was the introduction of a large number (N) of flavours [11–13] as a mathematical tool, employing the natural intuition that extra flavours do not introduce qualitatively new element to the theory except that the flavour symmetry group is enlarged. However, eventually, it was established beyond question that the $N \to \infty$ limit for finite-density fermions is not described by a mean-field theory [14, 15]. The physical reason behind this is the fact that there exists a large amoun if residual quantum-fluctuations of the Fermi surface. Following these seminal works, controlled approximations were folrmulate which were not based on expansion in $1/N$. One methodology was deforming the dynamics of the theory, for example by modifying the dispersion of a critical boson, to suppress quantum fluctuations at low-energies [16, 17]. In another effort, accessing two-dimensional NFLs was made possible by increasing the number of one-dimensional chains, where bosonization provides a controlled analytical tool [10]. Finally, the mathematical control of perturbation was gained by an intermediate step of modifying the number

of dimensions of spacetime (in which the system is embedded) in a continuous manner. This process can be implemented by extending either Fermi surface's dimension [**?**] or its co-dimension [18]. In this paper, we adopt the framework where a controlled approximation was devised via a dimensional regularization scheme, where the co-dimension of Fermi surface is extended to access the NFLs [1, 19]."

2. While the formalism is carefully developed, the manuscript is dense, and the readers who are not already familiar with the authors' prior papers may find it challenging to disentangle what is genuinely new from what is being reviewed.

We believe that we already wrote a detailed discussions in the last section of the new things reported in the paper, compared to earlier works. We repeat it here for the convenience of the referee:
"In this paper, we have analysed the nature of generic displacements of the critical Fermi surface for the NFL system arising at the Ising-nematic QCP, by using the quantum-Boltzmann-equation formalism applicable for our NFL scenario. By incorporating a generalized fermionic distribution function, we have derived the self-consistent equations whose solutions give us the functional forms of the Fermi-surface displacements as well as their dispersions. In particular, we have been focused on determining the dispersion and stability of the low-energy and low-momentum excitations/collective modes, arising from these displacements. Compared with our earlier work, we have included here the effects of damping, originating from the collision integral for the fermionic QBE. We have discussed the implications in two scenarios: (1) The simplistic F_0 model, where we are able to find an analytical solution by simply setting $F(\theta, \Omega) = F_0(\Omega)$, retaining only the $\ell = 0$ angular-momentum component. (2) The F_ℓ model, where generic $\ell \neq 0$ components are included, which warrants a numerical simulation to obtain the explicit results. A summary of our findings is elucidated below:

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We also refer the referee to the first paper in this series, viz. Ref. [4], where a clear derivation of the starting action in the Keldysh formalism is shown, starting with the equilibrium action the stable NFL fixed point at the Ising nematic QCP. In particular, Sec. II of Refs. [4, 5] contains all these details. All the above references have been properly cited in our manuscript and the origins of our action appropriately explained in Sec. II (entitled "Model") of our manuscript, without repeating every detail explained already in Refs. [4, 5]. Overall, we would like to emphasize the following:

- We have clarified in the Introduction (Sec. I) that the primary new advance of the present work is the systematic inclusion of *collision-induced damping* of collective modes within the QBE framework, going beyond the collisionless analysis of earlier works.
- We emphasize that the generalized Landau interaction functions appearing here are *frequency-dependent and complex*, encoding damping effects that qualitatively modify collective-mode dispersions and lifetimes.
- We highlight that the analysis applies to *generic angular-momentum channels* and finite $(\Omega, \mathbf{q})$, extending earlier treatments that focused primarily on collisionless dynamics.

The additions are intended to make the structure and novelty of the manuscript clearer, keeping in mind the readers who are not intimately familiar with our previous works.

I. LIST OF CHANGES

All additions have been marked in the colour red.

[1] D. Dalidovich and S.-S. Lee, Perturbative non-Fermi liquids from dimensional regularization, Phys. Rev. B **88**, 245106 (2013).
[2] I. Mandal and S.-S. Lee, Ultraviolet/infrared mixing in non-Fermi liquids, Phys. Rev. B **92**, 035141 (2015).
[3] S.-S. Lee, Recent developments in non-Fermi liquid theory, Ann. Rev. Condensed Matter Phys. **9**, 227 (2018).
[4] I. Mandal, Zero sound and plasmon modes for non-Fermi liquids, Physics Letters A **447**, 128292 (2022).
[5] K. R. Islam and I. Mandal, Generic deformation channels for critical fermi surfaces in the collisionless regime, Annals of Physics **457**, 169409 (2023).
[6] R. E. Prange and L. P. Kadanoff, Transport theory for electron-phonon interactions in metals, Phys. Rev. **134**, A566 (1964).
[7] Y. B. Kim, P. A. Lee, and X.-G. Wen, Quantum Boltzmann equation of composite fermions interacting with a gauge field, Phys. Rev. B **52**, 17275 (1995).
[8] S. Han, F. Desrochers, and Y. B. Kim, Fermi surface bosonization for non-Fermi liquids, Phys. Rev. B **112**, 165116 (2025).
[9] I. Esterlis, H. Guo, A. A. Patel, and S. Sachdev, Large-n theory of critical Fermi surfaces, Phys. Rev. B **103**, 235129 (2021).
[10] H.-C. Jiang, M. S. Block, R. V. Mishmash, J. R. Garrison, D. N. Sheng, O. I. Motrunich, and M. P. A. Fisher, Non-Fermi-liquid d-wave metal phase of strongly interacting electrons, Nature (London) **493**, 39 (2013).
[11] B. L. Altshuler, L. B. Ioffe, and A. J. Millis, Low-energy properties of fermions with singular interactions, Phys. Rev. B **50**, 14048 (1994).
[12] J. Polchinski, Low-energy dynamics of the spinon-gauge system, Nuclear Physics B **422**, 617 (1994).

[13] E.-A. Kim, M. J. Lawler, P. Oreto, S. Sachdev, E. Fradkin, and S. A. Kivelson, Theory of the nodal nematic quantum phase transition in superconductors, Phys. Rev. B **77**, 184514 (2008).
[14] S.-S. Lee, Low-energy effective theory of Fermi surface coupled with U(1) gauge field in $2+1$ dimensions, Phys. Rev. B **80**, 165102 (2009).
[15] M. A. Metlitski and S. Sachdev, Quantum phase transitions of metals in two spatial dimensions. I. Ising-nematic order, Phys. Rev. B **82**, 075127 (2010).
[16] C. Nayak and F. Wilczek, Renormalization group approach to low temperature properties of a non-Fermi liquid metal, Nuclear Physics B **430**, 534 (1994).
[17] D. F. Mross, J. McGreevy, H. Liu, and T. Senthil, Controlled expansion for certain non-Fermi-liquid metals, Phys. Rev. B **82**, 045121 (2010).
[18] T. Senthil and R. Shankar, Fermi surfaces in general codimension and a new controlled nontrivial fixed point, Phys. Rev. Lett. **102**, 046406 (2009).
[19] I. Mandal, UV/IR mixing in non-Fermi liquids: Higher-loop corrections in different energy ranges, Eur. Phys. J. B **89**, 278 (2016).